Winston Li

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Education

UC Irvine • B.S. in Computer Science w/ Honors, 3.9 / 4.0 GPA — Sep 2024 – Jun 2026

- Graduated in 2 years • ICS Honors Program • UROP Fellow
- Data Structures • Algorithms • Software Eng. • Operating Systems • Compilers
- Cryptography • Formal Lang. & Automata • Quantum Computation • Distributed Systems

Projects

tatic • *Rust, WebAssembly, dependent type theory* — Mar 2026 – Present

- Engineering a **JIT** for a higher-order language, caching compiled **WebAssembly** by content hash.
- Gating installation on agreement with an interpreter and a **kernel-checked proof** of the compiler model; unprovable terms stay interpreted.
- Writing an independent checker replaying the specializer, and translation validation from the IR.
- Fuzzing the JIT against the interpreter: 1000 seeds × 12 trials, zero mismatches, planted bugs caught.
- Mining rewrite rules from laws the prover couldn't close; seeded libraries need 3–5× fewer machine proofs (one theory).

Lambast • *Honors research project using Rust* — Feb 2025 – Jun 2026

- Developed a whole-program optimizing compiler for untyped λ-calculus using **closure conversion** and **De Bruijn indices**.
- Generated randomized benchmark terms via Grygiel & Lescanne's binary lambda-term enumeration; presented at the **UCI UROP Symposium**.

renno • *Rust*

- Designing a scripting language on a **CEK machine** with multi-shot **algebraic effect handlers**.
- Adding **gradual typing**, effect rows, refinements, length-indexed vectors; 400+ tests.

cereal • *C99*

- Writing a C99/GNU front end in C99: preprocessor, macro lint, parser, checker, **language server**.
- Matching gcc-13 diagnostics on 3,908/3,910 runnable gcc.dg tests; fuzzing parallel `-E` for same output.

Skills and interests

Proficient in Rust, C / C++, JS / TS, Python, Svelte, Tailwind.

Familiar with Git, HTML / CSS, Nix, LaTeX, Typst, Claude Code, Unix systems & shells.

Acquainted with most mainstream languages, e.g., C#, Go, Haskell, Java, Lua, Ruby, SQL.

Research interests ≈ programming language design & theory.

Experience

Learning Assistant • UC Irvine — Mar 2025 – Jun 2026

- Delivered supplementary pedagogy for **Python** and upper-div **Algorithms,** e.g., Master Theorem.
- Spearheaded study sessions to identify and guide students through gaps in understanding.

Intern • LGBTQIA+ Identity Commission @ UC Irvine — Oct 2024 – Jun 2025

- Led a team of 3 to organize and host themed Gayme Show™.

Programmer • Video Game Development Club @ UC Irvine — Sep 2024 – May 2025

- Worked with team of 13 to create typing game in **C#** and **Unity**; led dev of the main UI window.